

COMPUTER NETWORKS

Week-1

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Division of Marks

Quiz	15
Assignments	10
Mid Term Tests (2)	30
Final Exam	45
Total	<u>100</u>

Recommended Books

- Computer Networking, A top-Down Approach, James F. Kurose and Keith W. Ross, 8th Edition
- Computer Networks and Internets, Douglas E. Comer, 6th Edition
- Soft copies will be posted in Google Classroom

Important Topics to be covered

- Intro to the concept of networking
- Internetworking
- Network standards
- ISO & OSI (The OSI layering model → the Open Systems Interconnection Model)
- Understanding internet
- Web protocols
- Transport vs Network layers
- Congestion control, avoidance and recovery

Topics - Contd.

- Addressing schemes
- IPv4 → IPv6
- Routing algorithms
- Network Management protocols
- Wireless networks
- Cellular networks
- Latest trends in computer networks

Why Networks? -1

- Every app you love—Instagram, Google, Netflix—works because of **networks you cannot see**
- By the end of this course, you'll understand what's happening behind the screen”

Why Networks? -2

- A living Network
- Nervous system → neurons + synapses = packet-switched signals
- Blood circulation → arteries, veins = high-capacity links
- Hormonal system → slow, delayed signaling (high latency, low bandwidth)
- If routing fails in your nervous system → paralysis
- If congestion happens in blood vessels → heart attack
- Synapses: allows neurons to communicate with each other and integrate information from different regions of the brain

Why Networks? -3

- Ant Colonies & Bee Hives
 - No central server. No boss.
 - Ants communicate via pheromone trails
 - Bees share info through waggle dance
- Ants solved distributed systems long before humans invented routers

Pheromone trails

- when ants find food, they secrete an invisible chemical called "trail pheromone" as they return to the nest
- This trail leads directly from the nest to the food source
- Other worker ants then follow the trail right to the food
- Each worker then reinforces the trail on the way back

Waggle Dance

- A communication behavior in honeybees
- A dancing bee moves in a straight line and then in a semicircle to convey information about the distance and direction of a food source to other colony members

Why Networks? -4

- Cities & Networks
- Road networks → congestion, bottlenecks
- Electric grids → cascading failures
- Water supply → pressure & flow control
- Gas pipelines → secure transmission

Can a city survive if even one of these networks collapses?

Why Networks? -5

- Financial Systems
 - ATM → Bank → Clearing house → Bank
 - Credit card authorization happens in milliseconds
 - Stock trading relies on microsecond latency
- Your money doesn't move. Messages about your money move

Why Networks? -6

- Social networks
- Postal system
- Trade routes (Silk Road)
- Telegraph networks
- The Internet didn't invent networking. It just made it faster and global

Why Networks? -7

- Digital Life
- One minute without Networks?
- No mobile network
- No internet
- No GPS
- No banking
- No emergency services
- Is this still modern life—or survival mode?
- Wherever there is life, coordination, or intelligence → there is a network

Computer Network

- ♦ **Network**

System for connecting computers using a transmission technology

The computers can communicate with each other by sending and receiving data from each other

- ♦ **Internet**

Set of networks connected by routers that are configured to pass traffic among any computers attached to any network in the set

Life Today..

- Networks are part of :
 - Businesses
 - Homes
 - Government
 - Education field

Computer Networks ...growing ...

- Internet has grown into a communication system that reaches all countries of the world
- Many users have high speed Internet access through cable modems, DSL, optical fiber, or wireless technologies

Internet?

- Initially, these devices were desktop computers along with servers to transmit data and information in the form of web pages and emails
- Today, we see smart phones, laptops, tablets etc.
- **Additionally**, internet today includes TVs, gaming consoles, home security systems, home appliances, watches, eye glasses, cars, traffic control systems,, much more

Internet..

- Now we do not have ONLY computer networks, rather we have IoT and IoE
- IoT = Network + things (any physical object that generates data)
- IoE = Network + things + people + data + processes
- Is there anything like IIoT ?

Internet..

- Now we do not have ONLY computer networks, rather we have IoT and IoE
- IoT = Network + things (any physical object that generates data)
- IoE = Network + things + people + data + processes
- **Is there anything like IIoT ?**
- The Industrial Internet of Things (IIoT) → A network of intelligent devices connected to form systems that monitor, collect, exchange and analyze data
- Number of devices connected to the Internet in 2030 is expected to reach 50 billion compared to approximate 8 billion population of the world today

End Systems

- End systems are connected via a network of **communication links** and **packet switching**
- Communication links are made up of different types of physical media, including coaxial cable, copper wire, optical fiber, and radio spectrum
- Different links can transmit data at different transmission rates

Internet Service Providers

- End systems (computers, servers, mobile devices etc.) access the Internet through **Internet Service Providers (ISPs)**
- These include residential ISPs such as local cable or telephone companies, corporate ISPs, university ISPs, ISPs that provide WiFi access in airports, hotels, coffee shops, and other public places, and cellular data ISPs, providing mobile access to our smartphones and other devices
- ISPs that provide access to end systems must also be interconnected

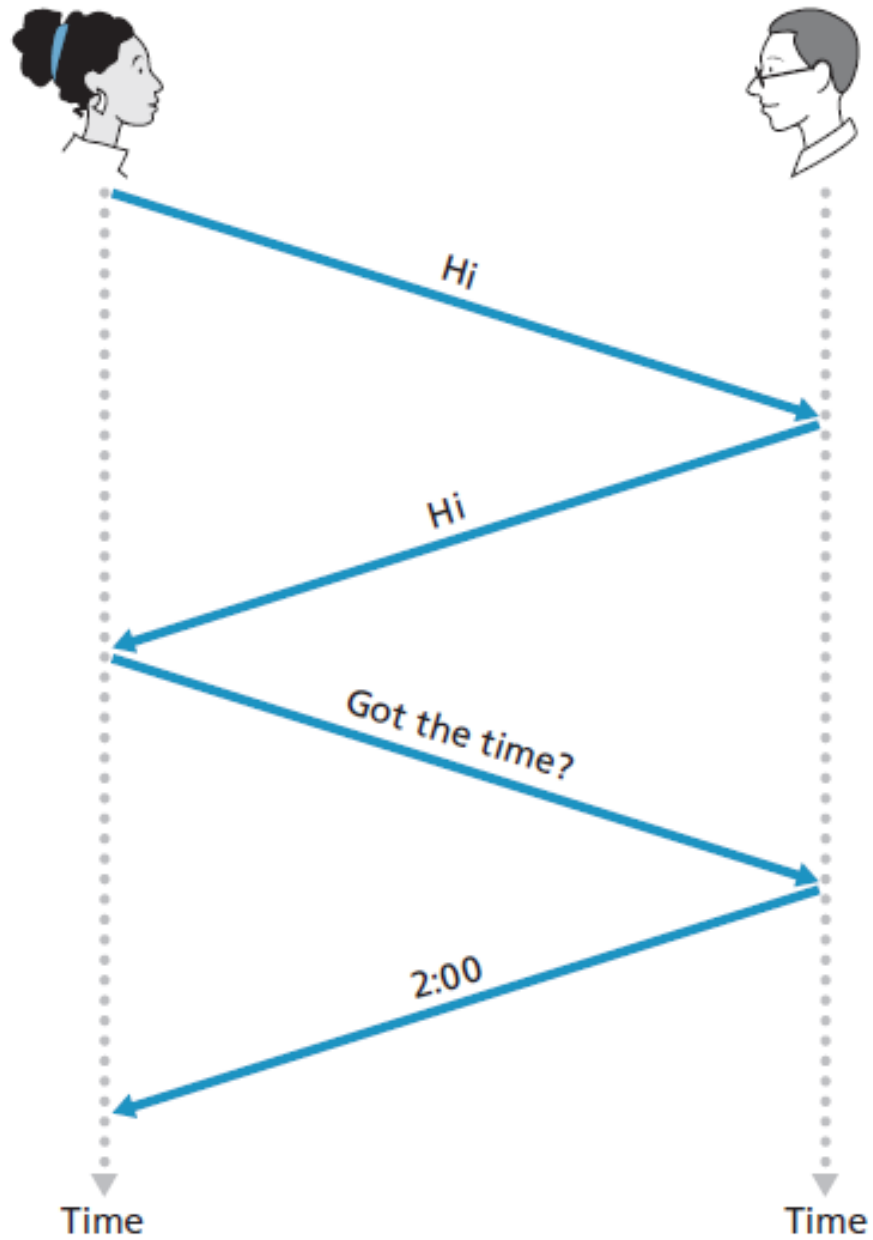
Protocols

- **Protocols** control the sending and receiving of information within the Internet
- The **Transmission Control Protocol (TCP)** and the **Internet Protocol (IP)** are two main protocols in the Internet
- The IP protocol specifies the format of the packets that are sent and received among routers and end systems
- It is the set of rules governing the format of data sent via the internet or local network

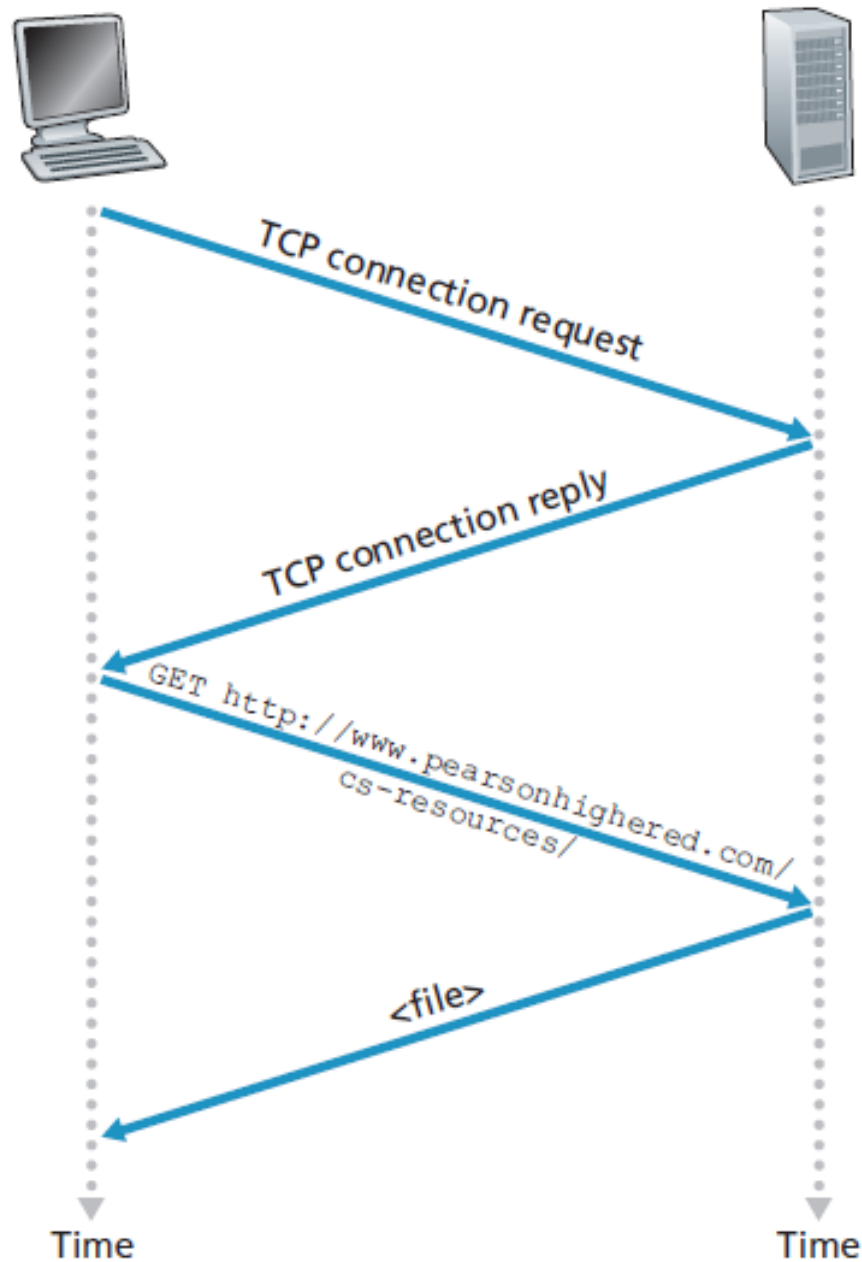
Protocols

- **IP** is the part that obtains the address to which data is sent
- **TCP** is responsible for data delivery once that **IP** address has been found
- Everyone must agree on what each and every protocol does, so that people can create systems and products that interoperate

Human Protocol



Network Protocol



Protocols

- In human protocols, **Hi** is one way of replying
- Other person could have replied otherwise ...
- In our human protocol, there are specific messages we send, and specific actions we take in response to the received responses
- If one understands the concept of time and the other does not, the protocol won't interoperate and no useful work can be accomplished
- So, in networking—it takes two (or more) communicating entities running the same protocol in order to accomplish a task

Network Protocols

- In a network protocol, the entities exchanging messages and taking actions are hardware or software components
- for example, computer, smartphone, tablet, router, or other network-capable device
- Congestion-control protocols in end systems control the rate at which packets are transmitted between sender and receiver
- Protocols in routers determine a packet's path from source to destination

Types of networks by Size

- Local Area Networks (LANs)
- E.g. home WiFi network, a business office, group of buildings
- Wide Area Networks (WANs)
- E.g. Internet

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Network Classification

- A LAN, network occupies the smaller area like a room a floor or a building (also have WLANs)
- In WAN, network occupies larger areas like cities & countries
- Internet is a Wide Area Network
- LAN & WAN are compared by the speed of transmission, bandwidth, security, reliability, management and standards etc.

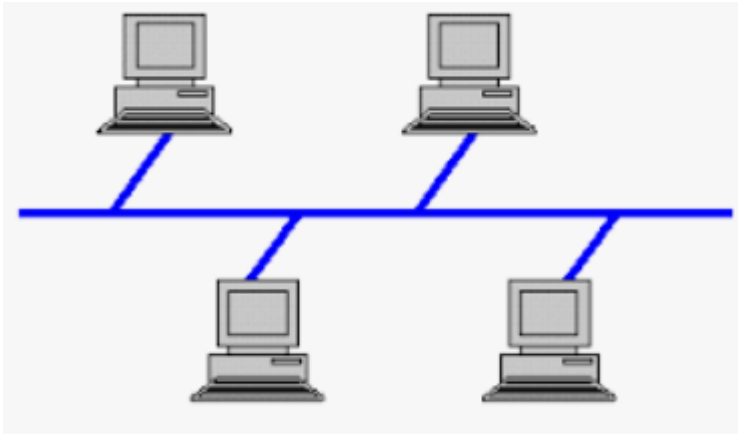
MAN

- Metropolitan Area Network (MAN)
 - covers larger area by connecting LANs
 - The size of the Metropolitan area network is larger than LANs and smaller than WANs(wide area networks), a MAN covers the larger area of a city or town
 - E.g. MAN where libraries of all universities of Lahore are connected
-
- Read about other types of networks

Topology

- Physical and logical layout of the network
- Physical layout refers to the connected devices and nodes
- The logical topology focuses on the pattern of data transfer between network nodes

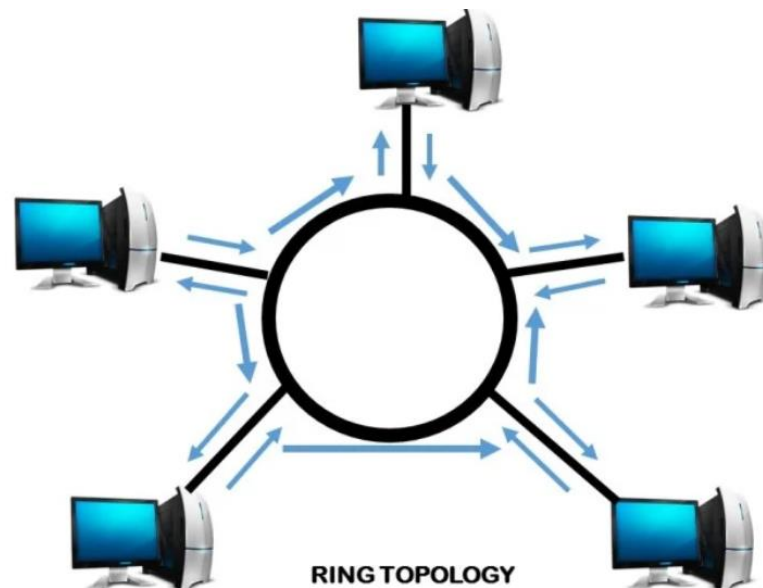
Bus Topology



- A *bus topology* is a *topology* for a *Local Area Network (LAN)* in which all the nodes are connected to a single cable. The cable to which the nodes connect is called a "backbone"
- If the backbone is broken, the entire segment fails.

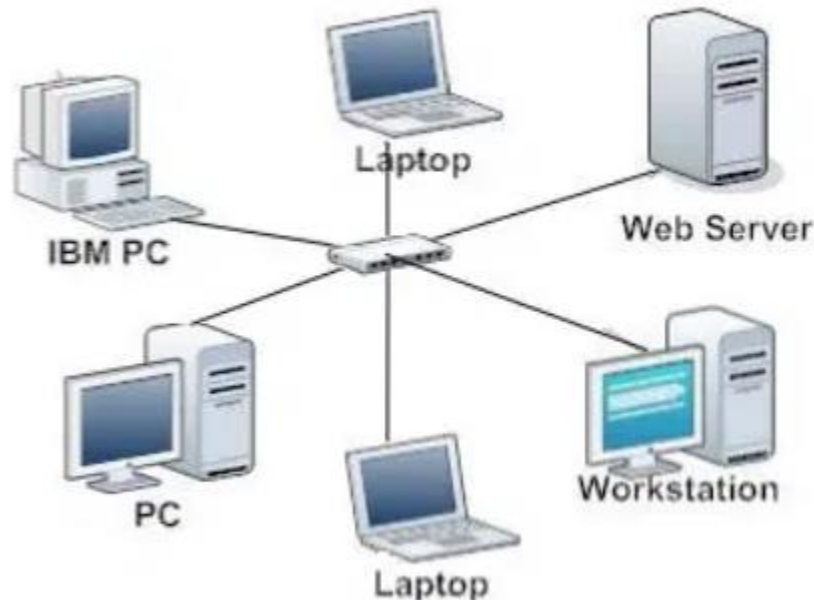
Ring topology

- Device connections create a circular data path
- Each networked device is connected to two others, like points on a circle



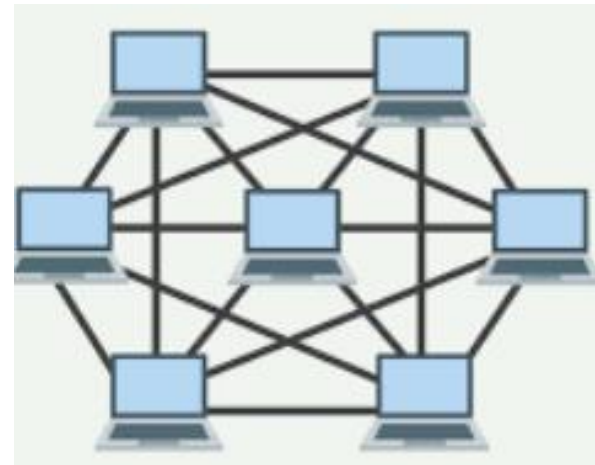
Star Topology

- A LAN topology in which all nodes are individually connected to a central connection point, like a hub or a switch
- The central hub is usually a fast, self contained computer and is responsible for routing all traffic to other nodes

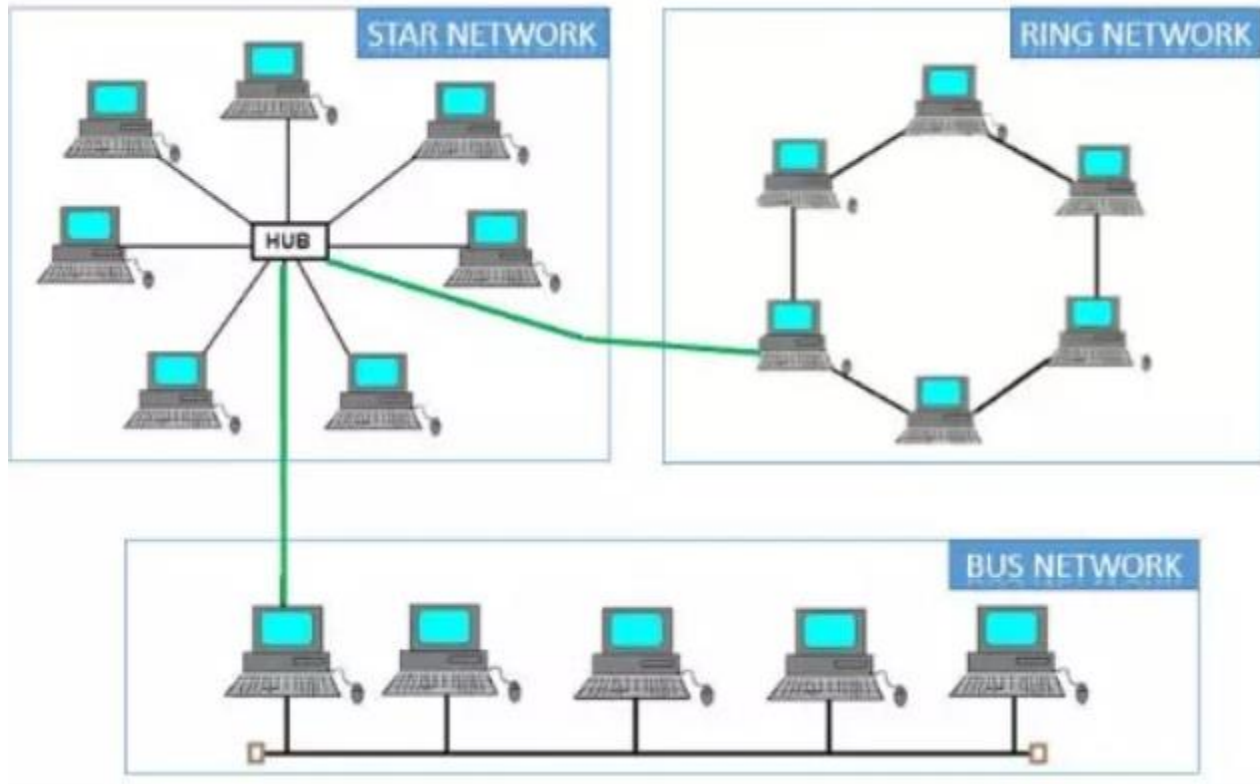


Mesh Topology

- There is no central connection point
- Each node is connected to at least one other node and usually to more than one
- Each node is capable of sending messages to and receiving messages from other nodes
- The nodes act as relays, passing on a message towards its final destination



Hybrid Topology



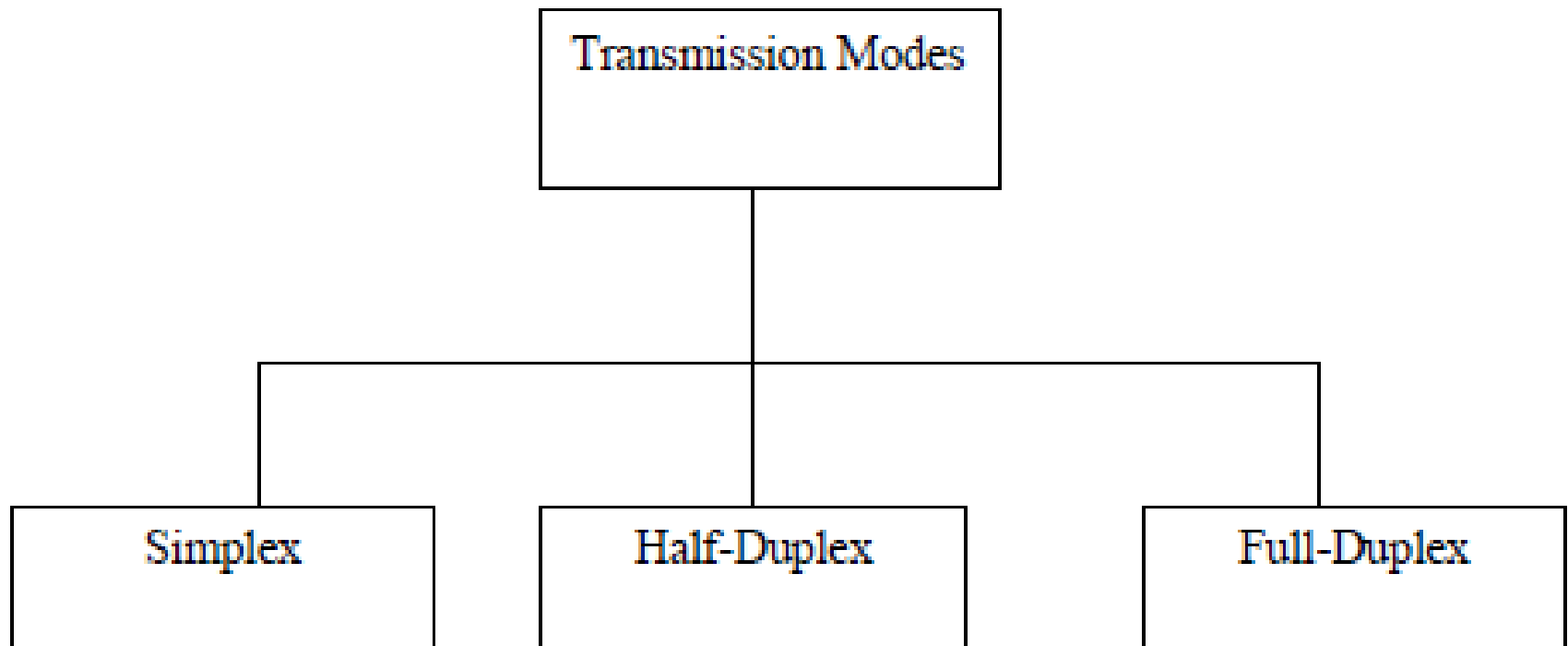
Hybrid Topology

- Star--Ring and Star--Bus networks are most common examples of **hybrid network**
- The star bus **topology** is combined with the linear bus in large networks
- In such cases the linear bus provides a backbone that connects multiple stars

Transmission Mode

- “Transmission Mode is used to define the direction of the signal flow between the linked devices”

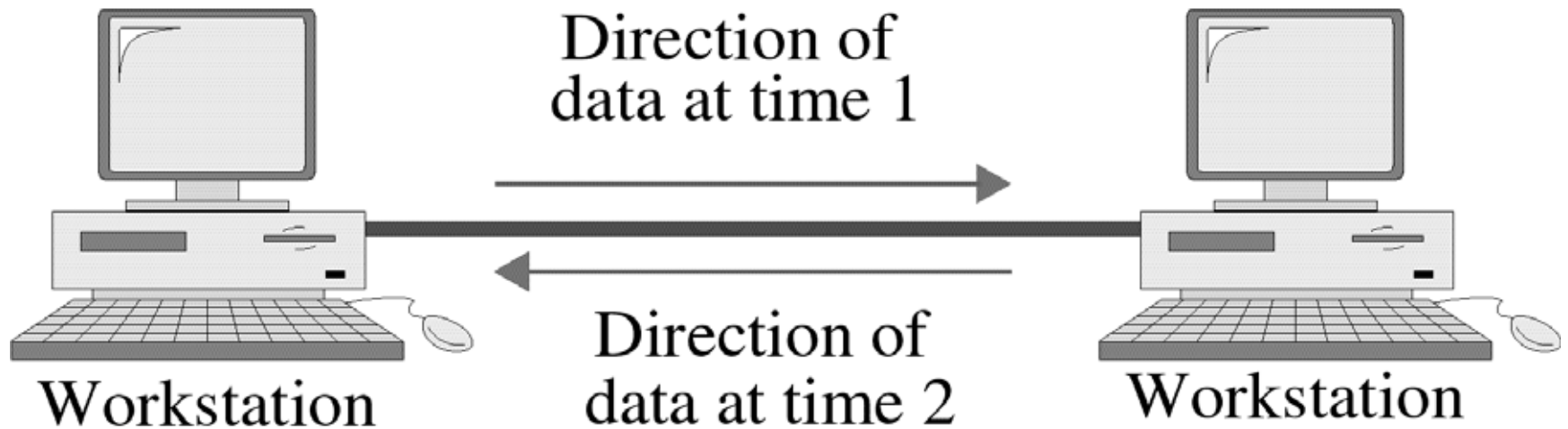
Transmission Mode



Simplex Mode

- Communication is Unidirectional
- Only one of the two stations can transmit
- Other can only receive
- E.g. Radio, TV
- They receive data from stations but cannot transmit the data back
- data transmission from computer to printer
- From keyboard, mouse, joystick to computer
- Loud speaker
- Fire alarm system
- TV and remote

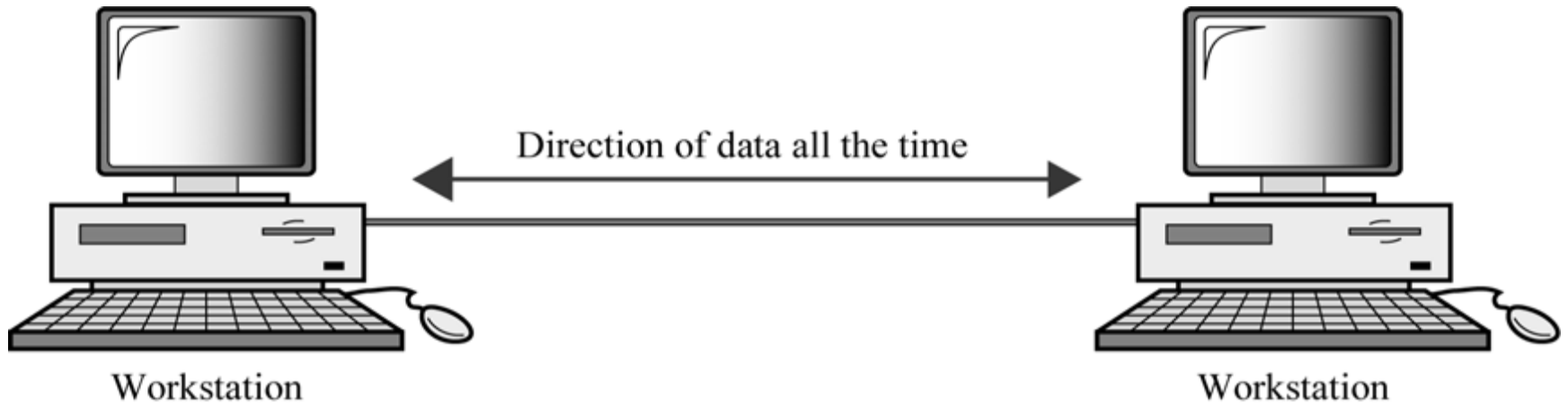
Half - Duplex Mode



Half-Duplex Mode

- Each station can both transmit and receive but not at the same time
- When one device is sending the other can only receive and vice versa
- For example:
- One Lane Road with two directional traffic
- Internet surfing
- Full Channel capacity is allocated to whatever entity that is transmitting at a specific time
- Walkie Talkies

Full Duplex (Duplex)



Full Duplex (Duplex)

- Both stations can transmit and receive simultaneously
- Two way street with traffic flowing in both directions at the same time
- E.g. Audio calls on telephones. You can talk and hear at the same time\
- Remote desktop connection (send and receive data from one system/PC to another PC on the same network.

Full Duplex (Duplex)

- The sharing can take place in two ways:
- Either the link must contain two physically separate transmission paths:
 - One for sending and
 - One for receiving
- Capacity of the channel is divided between signals traveling in opposite directions
 - EXAMPLE
 - Telephone Networks
 - When two people are communicating via a telephone line, both can talk and listen at the same time

Transmission Modes

- A wired Ethernet network is **full duplex**, meaning a device can send and receive, or upload and download, simultaneously
- **WiFi** is half **duplex**, so if a client is sending data to the AP(Access Points), the AP can not also send data to the same or any other client at the same time
- For example if you are downloading a file and sending an email at the same time , these two processes will happen in turns but the lag goes unnoticed because time difference is very little

Internet Domain Names

- Internet address names are strictly controlled
- → so no one could add a computer to the Internet that had the same address as another computer
- Address name has two parts → computer name and its domain
- [computer.domain](#)
- Some computer names have several parts separated by periods, so some addresses have the format [computer.computer.computer.domain](#)
- For example, the main university [www.bnu.edu.pk](#)
- Web server for the Kelley School of Business at Indiana university is [www.kelley.indiana.edu](#)

Internet Domain Names

1. Internet began in the United States, the American address board was the first to assign domain names
2. Some common U.S. domain names are as follows:

EDU	for an educational institution, usually a university
COM	for a commercial business
GOV	for a government department or agency
MIL	for a military unit
ORG	for a nonprofit organization

Internet Domain Names

- As networks in other countries were connected to the Internet, they were assigned their own domain names

CA	for Canada
AU	for Australia
UK	for the United Kingdom
DE	for Germany

Internet Domain Names

- New top-level domains that focus on specific types of businesses continue to be introduced

AERO	for aerospace companies
MUSEUM	for museums
NAME	for individuals
PRO	for professionals, such as accountants and lawyers
BIZ	for businesses

Components of Networks

- The **server** stores data or software that can be accessed by the clients
- In client-server computing, several servers may work together over the network with a client computer to support the business application
- The **client** is the input-output hardware device at the user's end of a communication circuit
- It provides users with access to the network and the data and software on the server

Components of Networks

- The **circuit** is the pathway through which the messages travel → a copper wire, fiber-optic cable or even wireless transmission
- Many devices in the circuit perform special functions such as switches and routers
- A network does not need a server
- Some networks are designed to connect a set of similar computers that share their data and software with each other → **peer to peer networks**
- The computers function as equals, rather than relying on a central server

Types of networks

- Two other common terms are **intranets** and **extranets**
- An **intranet** is a LAN that uses the same technologies as the Internet (e.g., Web servers, Java, HTML [Hypertext Markup Language])
- It is open to only those inside the organization
- For example, some pages on a Web server may be open to the public and accessible by anyone on the Internet
- Some pages may be on an intranet and therefore hidden from those who connect to the Web server from Internet
- Sometimes, an intranet is provided by a completely separate Web server hidden from the Internet
- E.g. an intranet for the IT Department at BNU with information on faculty expense budgets, class scheduling in the current semester

Types of networks

- An **extranet** also uses the same technologies as the Internet
- It is open for users outside the organization who access it over the Internet
- It can provide access to information services, inventories, and other internal organizational databases that are provided only to customers, suppliers, or those who have paid for access
- Could be done through passwords, smart cards
- Many universities provide extranets for Web-based courses so that only those students enrolled in the course can access course materials and discussions

Network standards

- **Standards** are necessary in almost every business and public service entity
- The transmission of electric current was not standardized until the end of the nineteenth century
- Customers had to choose between Thomas Edison's direct current (DC) and George Westinghouse's alternating current (AC)
- The primary reason for standards is to ensure that hardware and software produced by different vendors can work together
- Without networking standards, it would be difficult to share information
- Standards → customers are not locked into one vendor
- They can buy hardware and software from any vendor whose equipment meets the standard
- The use of standards makes it much easier to develop software and hardware that link different networks

ISO?

- One of the most important standards-making bodies

International Organization for Standardization

- Makes technical recommendations about data communication interfaces (see www.iso.org)
- ISO is based in Geneva, Switzerland. The membership is composed of the national standards organizations of each ISO member country

Keeping Up with Technology

- The data communications and networking arena changes rapidly
- It is therefore important for network managers to keep up with these changes
- For a network to operate, many different standards must be used simultaneously
- Either the sender and receiver of a message must use the same standards or, more likely, there are devices between the two that translate from one standard into another
- Networks often use software and hardware designed for different standards → a lot of translation is done

Wireless LAN and BYOD

- Mobile devices, such as smartphones and tablets require employers to allow their employees to bring these devices to work and use them to access data, such as their work email
- This movement is called bring your own device, or **Bring Your On Device (BYOD)**
- Employers need to add or expand their Wireless Local Area Networks (WLANs) to support all these new devices

Security

- Employees can access not only their email but also other critical company assets, such as information about their clients, suppliers, or sales
- Employers need to be careful on how to manage access to company applications for BYOD
- Companies can adopt two main approaches: (1) native apps or (2) browser-based technologies
- **Native apps** require an app to be developed for each application that an employee might be using for every potential device that the employee might use (e.g., iPhone, Android, Windows)
- The **browser based** approach (often referred to as responsive design using HTML5) doesn't create an app but rather requires employees to access the application through a Web browser
- Both these approaches have their pros and cons

Massively Online

- Massively multiplayer online games, such as World of Warcraft, where you can play with thousands of players in real time
- Today Education is also massively online. Edx, Khan Academy, Lynda.com, or Code Academy have websites that offer thousands of education modules for children and adults to help them learn
- Massive open online courses, or MOOC. MOOC enable students who otherwise wouldn't have access to elite universities to get access to top knowledge without having to pay the tuition
- These classes are offered by universities, such as Stanford, UC Berkeley, MIT, UCLA, Carnegie Mellon free of charge and for no credit (although at some universities, you can pay and get credit toward your degree)

Massively Online

- Politics has also moved massively online
- Candidates reach out to the crowds and ordinary voters not only through apps like Facebook but also through Reddit and Google Hangouts
- President Trump's use of Twitter is unprecedented. He can directly reach millions of
- Finally, massively online allows activists to reach masses of people in a very short period of time

Future need

- Not only engineers who deeply understand the technical aspects of networks
- Also highly social individuals who embrace technology in creative ways to allow business to achieve a competitive edge
- So You are in the right place at the right time!

Book Ref

- Chp 1